Module 4 Narrative – Artifact 2 enhancement

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CS-499 – Computer Science Capstone

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February 1, 2026

This artifact is the Weight Tracker App. The enhancement is to add a search and sort feature. This will not be using sort capabilities in the database but instead will use in-memory sorting using a merge sort. The Weight Tracker app allows a user on android set a goal of weight loss or gain, add weights on various dates, then alert the user when the goal is reached. The app also has SMS alerts for when the goal is reached. This artifact was created in October 2025 as the main project in CS-360, mobile architecture and development.

This artifact encompasses several skills acquired in the computer science program. It produces a complete, usable, practical product. The enhancements make it into a full cloud-based application. It allows me to demonstrate my skills in application design, java programming, cloud computing and android development.

In this enhancement, the artifact was improved by adding a local sort, and search feature. Users can search by date or weight, and sort ascending or descending with a single tap on the date or weight fields.

I have achieved the course outcomes that I planned to meet. This enhancement, in my assessment is complete for this phase. As-is, the local SQLite database is used and the data is queried and then sorted locally. A search feature has been added which acts as a filter by ranges of date and/or weight. For enhancement three the database will be changed to Firebase but this should not require significant changes to the local sort algorithm.

This enhancement fulfills and exceeds the planned outcome in module one. I have achieved local sort, search and have prepared the app for a future firebase database backend.

Enhancing and modifying the artifact was not so straightforward. I had to make decisions as to which algorithm to use. I decided on merge sort because it is stable and consistent. It has an

O (n log n) time complexity. It is a classic divide and conquer strategy. Getting some of the ascending and descending sort to work properly took trial and error. I had to make UI decisions for both portrait and landscape. I coded both because Google's UI design guidelines for Android since Android 16 now require both landscape and portrait. (Choudhary, 2025)

Works Cited

Choudhary, S. (2025, September 3). *Android 16 update: No more portrait-only apps*. Medium. https://chsatyam.medium.com/android-16-update-no-more-portrait-only-apps-ee483b7b8703